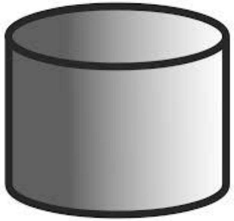
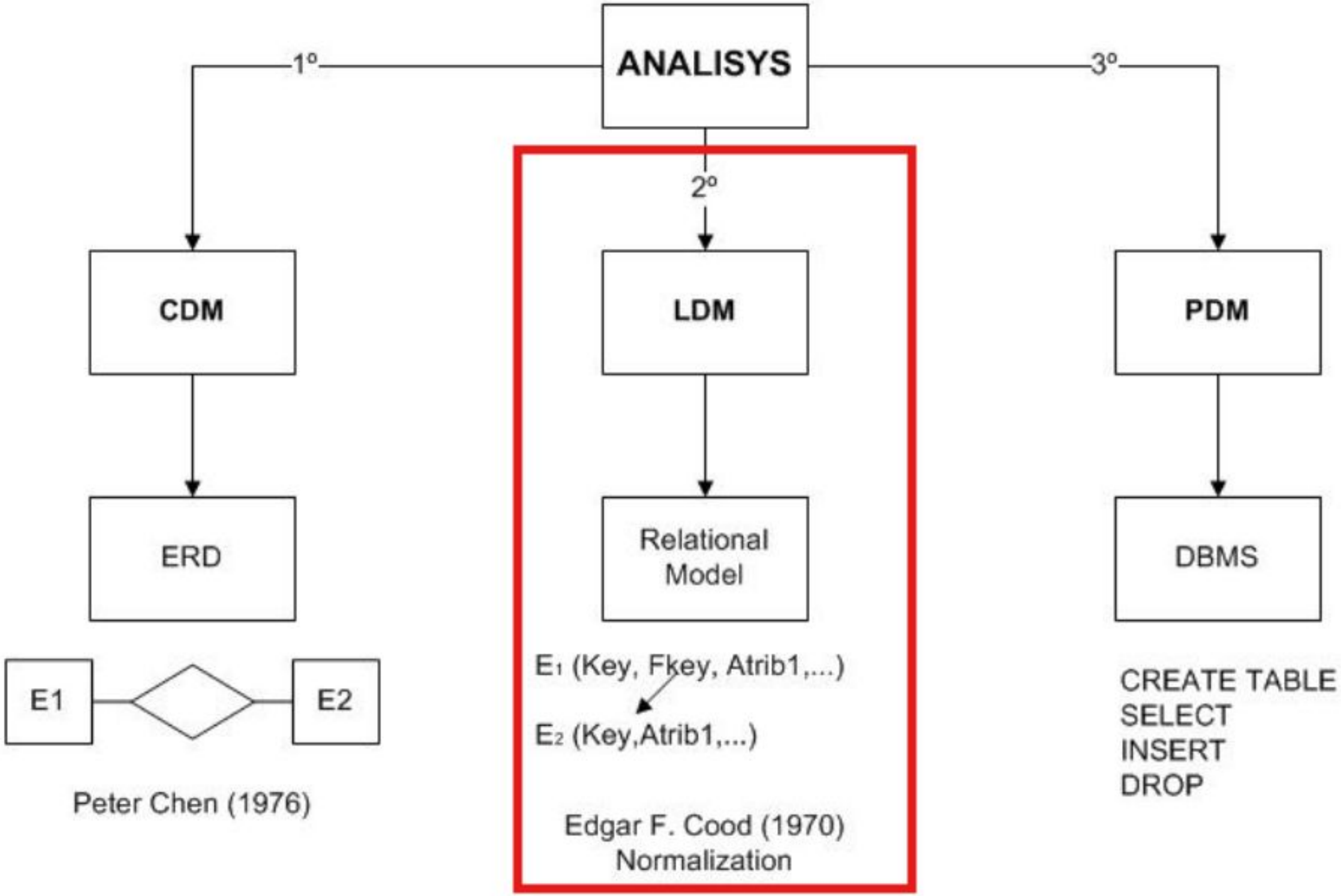
DATABASES

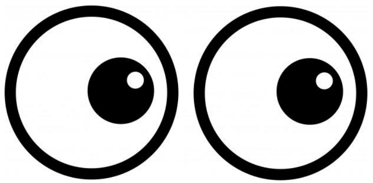


Unit 2 Database Design


ANALISYS
1º
2º
3º
CDM
LDM
PDM
ERD
Relational Model
DBMS
E1
E2
Peter Chen (1976)
E₁ (Key, Fkey, Atrib1,...)
E₂ (Key,Atrib1,...)
Edgar F. Cood (1970)
Normalization
CREATE TABLE
SELECT
INSERT
DROP

Index UNIT 2

1. Conceptual Data Model (2-1)
2. Logical Data Model (2-2)
3. Physical Data Model (3)

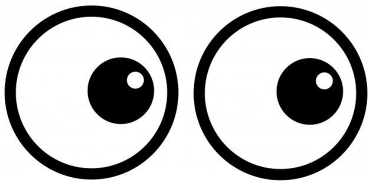


Index UNIT 2-1 Conceptual Data Model

1. Data modelling
2. Entity Relationship Diagram (ERD)
    a. Entities
    b. Attributes
    c. Relationships
3. Extended Entity-Relationship (EE-R) Diagram
    a. Superclass
    b. Subclass
    c. Hierarchical relationships

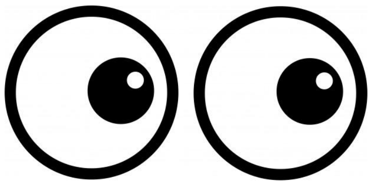
Index UNIT 2 - 2 (Logical Data Model)

1. From ERD to Relational Model
   a. The Relational Model
   b. Rules
   c. Entity to table
   d. Transforming Relationships
2. Normalization



Index UNIT 3 (Physical Data Model)

1. Relational Databases Definition
2. Data model
3. Tables
4. Columns
5. Rows
6. Keys
7. Relationships
8. Normalization
9. **SQL**
10. ACID Properties



From ERD to Relational Model

- Once our Entity/Relationship Diagram model (Conceptual Data Model) is finalised, it is time to convert it into tables (Relational Model) and relationships so that it can be implemented in our DBMS.

- The normalization model may not be the best in performance.

The Relational Model

The relational model

Defined in 1970 by Edgar F. Codd at IBM Research Center (San José, CA). Its main element is the **relation**: a bidimensional structure that represents a group of data objects. Each data object, constituting a row, is a **tuple**, and attributes go by the name **attribute**.

Within the database scope, relations are named **tables**, tuples **rows** or **records** and attributes keep their original name.

attribute

value

tuple (row = record)

ID	Name	Surname	Phone no.	Birthdate	Ins. date
007895211	Elizabeth	Jones	2036785241	1983/09/13	2012/09/20
506874522	Julian	Chapman	2039876455	1963/02/05	2011/09/14
385469983	Ian	McAlister	2051123542	1978/10/24	2009/05/05
096538014	Ruth	Hutchinson	2038900017	1981/05/15	2013/09/04

The Relational Model. Table

- A table is a set of related data that store information with a specific meaning within a database.
- Name of the table→ Name of the Entity or Relationship.
- Name of columns → Attributes of the Entity or Relationship.
- Row, record or tuple → A set of values, each belonging to the domain of the attribute it represents, that uniquely identify an element.
    - **There cannot be repeated rows in a table.**

Table

Table: Student

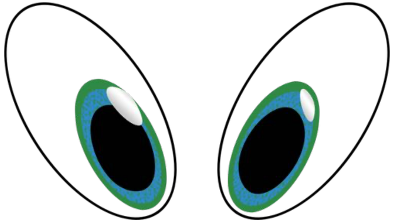
Attributes:

DNI	NAME	SURNAME
1	Elizabeth	Jones
2	Julian	Chapman
3	Ian	McAlister
4	Ruth	Hutchinson
5	Carless	Sanz

Record

The Relational Model. Rules

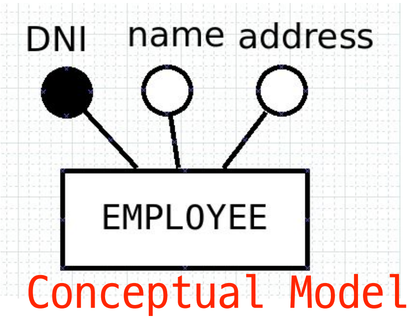
- The transformation of components of the entity-relationship diagram into elements of the physical data model follows the following rules:
    - Every entity becomes a table.
    - Every attribute becomes a field.
    - The attributes marked as part of the primary key are converted into primary key fields of the new table. Relationships present a casuistic based on their cardinality.



- **Remember**:
    - Primary key.
    - Candidate keys.
    - **Foreign key**. Set of attributes of an entity that are primary keys in another entity.

The Relational Model. Entity to table

- Each entity is transformed into a table.
- The primary key is underlined.



Logical Model

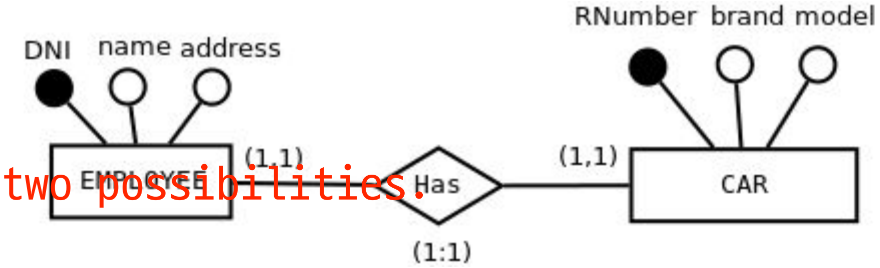
EMPLOYEE(<u>DNI</u>, name, address)

1:1

The Relational Model. Transforming Relationships

1:1 Relationships

- They do not generate a table.
- The key of one entity is inserted as a foreign key in the other entity (it is also possible to insert both keys as foreign keys, depending on the desired performance).



In 1:1, there are two possibilities.
Possibility 1.
EMPLOYEE to have CAR foreign key

EMPLOYEE(<u>DNI</u>, name, address, **RNumber**)

foreign key from CAR

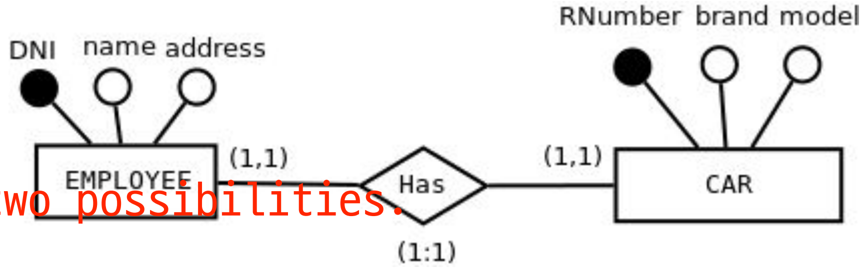
Use a circle to show it is a foreign key

CAR(<u>RNumber</u>, brand, model)

The Relational Model. Transforming Relationships

1:1 Relationships

- They do not generate a table.
- The key of one entity is inserted as a foreign key in the other entity (it is also possible to insert both keys as foreign keys, depending on the desired performance).



In 1:1, there are two possibilities.
Possibility 2.
CAR to have EMPLOYEE foreign key

EMPLOYEE(<u>DNI</u>, name, address)

CAR(<u>RNumber</u>, brand, model, **DNI**)

foreign key from EMPLOYEE

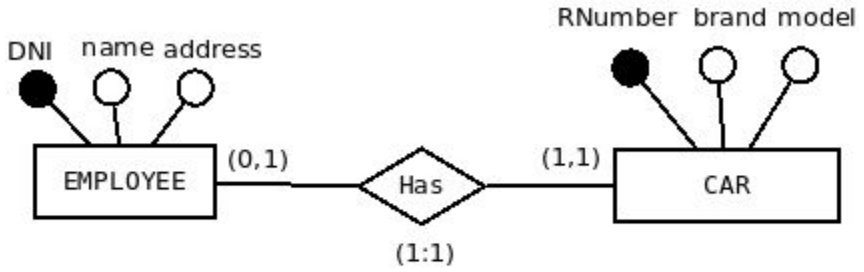
1:1 with one (0, 1)

The Relational Model. Transforming Relationships

1:1 Relationships

However, if there is a (0, 1) be careful where you put the foreign key

- They do not generate a table.
- If only one entity participate with (0,1) the key of the entity (1,1) is inserted as a foreign key in the other entity.



EMPLOYEE(<u>DNI</u>, name, address, **RNumber**)

foreign key from CAR

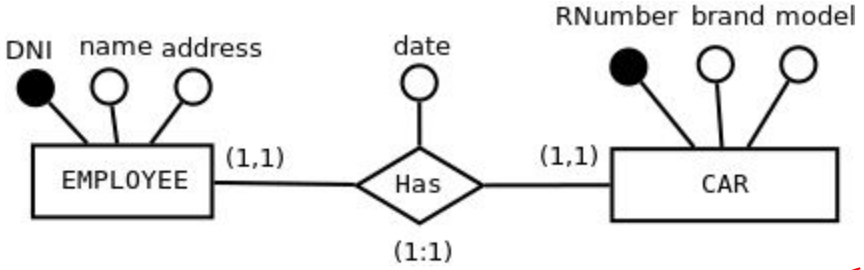
CAR(<u>RNumber</u>, brand, model)

1:1 with relation attributes

The Relational Model. Transforming Relationships

1:1 Relationships

- If the relationship has attributes they are passed to the table where the foreign key is entered.



date is a relationship attribute

EMPLOYEE(<u>DNI</u>, name, address, **RNumber,** date)

foreign key

CAR(<u>RNumber</u>, brand, model)

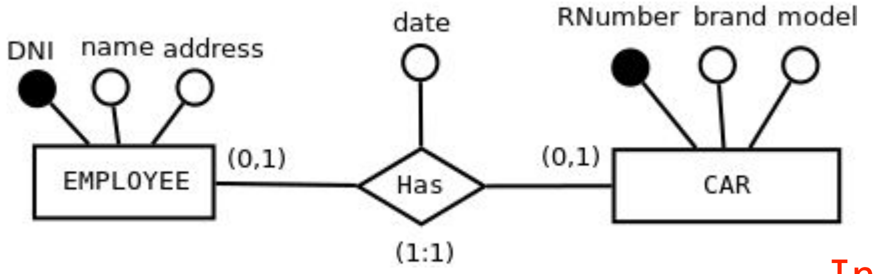
1:1 with two (0,1)

The Relational Model. Transforming Relationships

If we have 1:1 relationship,
but both can be (0, 1),
then we have to create three tables

1:1 Relationships

- If the two entities participate with (0,1) a new table is created for the relationship



EMPLOYEE(DNI, name, address)

CAR(RNumber, brand, model)

HAS(**DNI, RNumber**, date)

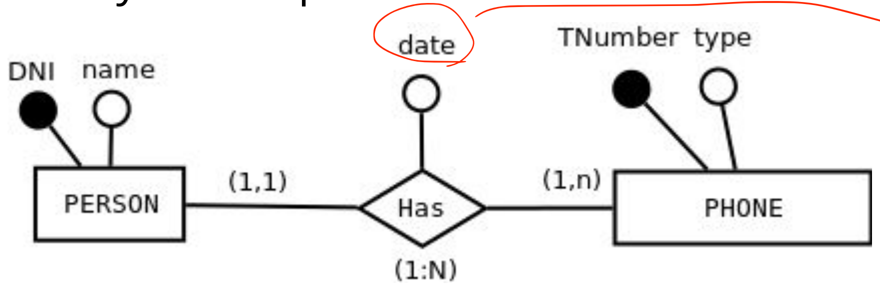
In table HAS, DNI and RNumber are together
composite, primary key
DNI is foreign key from EMPLOYEE
and RNumber is foreign key from CAR

1:N

The Relational Model. Transforming Relationships

1:N Relationships

- They don't usually generate a table.
- The key of the entity that participates with 1 (maximum) is passed as a foreign key to the table of the entity that participates with N.
- If the relationship has attributes, they are passed to the table where the foreign key is incorporated.



PERSON(DNI, name)

PHONE(TNumber, type, **DNI**, date)

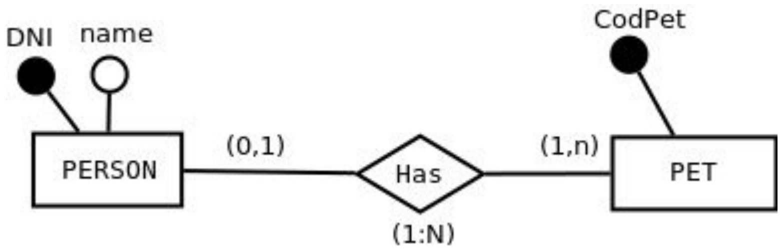
Person can have several phones, so add Person primary key to Phone
Phone has Person primary key as foreign key

1:N with (0, 1)

The Relational Model. Transforming Relationships

1:N Relationships

- Minimum participation zero.
- If it is on the N side nothing changes.
- If it is on the 1 side, a new table is created for the relationship with the key of both entities. **The key of the relationship will be only the key of the N entity**.



PERSON(<u>DNI</u>, name)

PET(<u>CodPet</u>)

HAS(<u>**CodPet**</u>, **DNI**)

In table HAS, only CodPet is primary key.
DNI is not a primary key bc person can be (0, 1), it is only foreign key

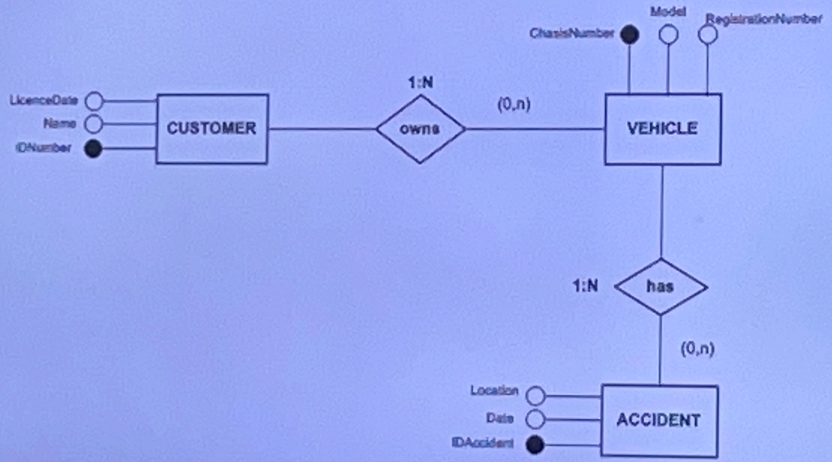
Exercises (III)

EXAM

1.- Build an E/R diagram and obtain the corresponding Relational Model for a car insurance company with a set of customers.

1. For each customer we know the ID number, name and date of obtaining a driving licence.
2. Each customer has a certain number of vehicles insured with the company.
3. For each vehicle we know the chassis number, model and registration number.
4. The same customer can have several vehicles insured.
5. For each vehicle, a number of accidents are registered for which information is also to be stored. For each accident we want to know the date of the accident and the location where it occurred.

Exercises (III). ERD

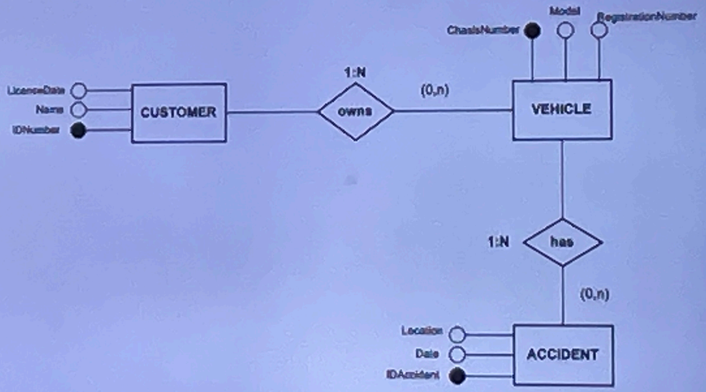


Exercises (III). Relational Model

CUSTOMER (IdNumber, Name, LicenceDate)

VEHICLE (ChasisNumber, Model, numBastidor, **IdNumber**)

ACCIDENTE (IDAccident, Date, Location, **ChasisNumber**)

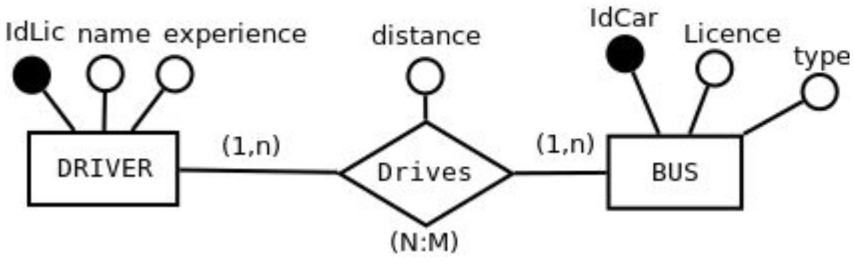


N:M | N:M with 0

The Relational Model. Transforming Relationships

N:M (N:N) Relationships

- They always generate a new table.
- A table is created. Its key consists of the keys of the entities participating in the relationship.
- If there are own attributes, they are passed to the table of the relationship.
- If there are zero minimum shares, the result is exactly the same (subrogate).



DRIVER(<u>IDLic</u>, name, experience)

BUS(<u>IdCar,</u>Licence,type)

DRIVES(<u>**IDLic, IdCar**</u>, distance)

IDLic, IdCar is one primary key, composed of two attributes

DRIVES(idDrive, **IDLic, IdCar**, distance) if (0,n) and (0,n)

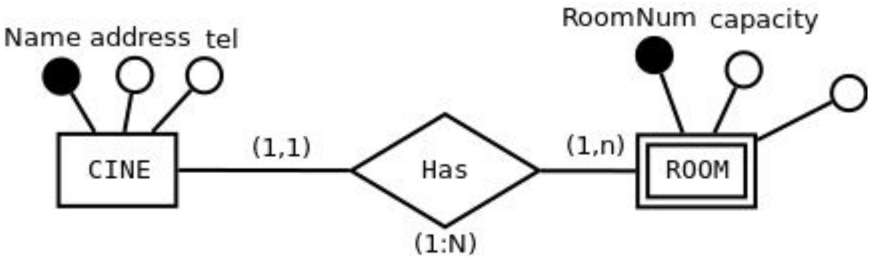
If there is 0, we have to invent/create a primary key(idDrive)

Weak and Strong Dependency Relationship

The Relational Model. Transforming Relationships

Dependency relationships in identification

- They do not usually generate a table because they are usually 1:1 or 1:N.
- The key of the strong entity must be entered in the table of the weak entity and be part of the key of the weak entity.



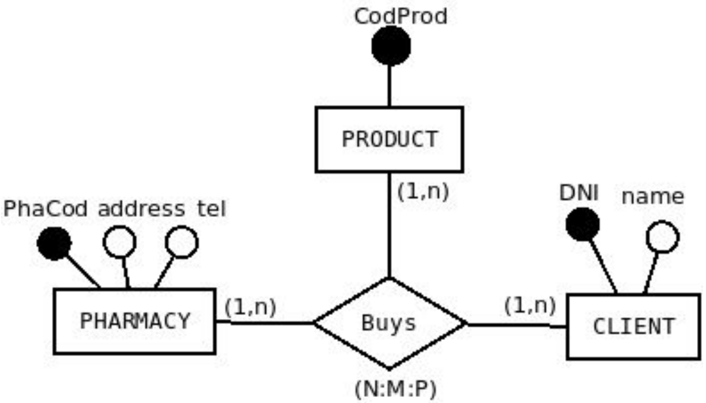
CINE(<u>Name</u>, address, tel)

ROOM(<u>**Name**, RoomNum,</u> capacity)

The Relational Model. Transforming Relationships

N-ary relationships

- They always generate a table.
- A new table is created with the keys of the entities participating in the relationship.
- The attributes of the relationship are incorporated into the table of the relationship itself.



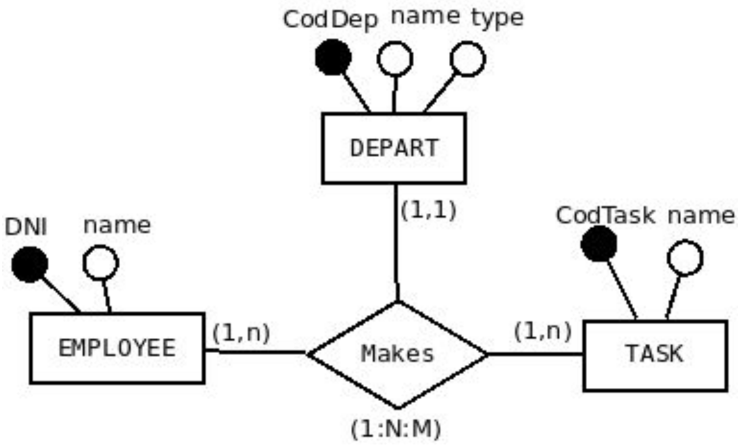
PHARMACY(PhaCod,address, tel)
CLIENT(DNI, name)
PRODUCT(CodProd)
BUYS(**PhaCod, dni, CodProd**)

Three attributes
create one primary key

The Relational Model. Transforming Relationships

N-ary relationships

- If an entity participates with maximum cardinality 1, it may not have to be part of the key of the table of the relationship



DEPART(<u>CodDep</u>,name, type)
EMPLOYEE(<u>DNI</u>, name)
TASK(<u>CodTask</u>, name)

NO → MAKES(<u>**DNI,CodTask,CodDep**</u>) ✗
BUT → MAKES(<u>**DNI,CodTask,**</u>**CodDep**) ○

However, if there is (1:N:M),
primary key of MAKES will be only DNI and CodTask,
and CodDep will be only foreign key

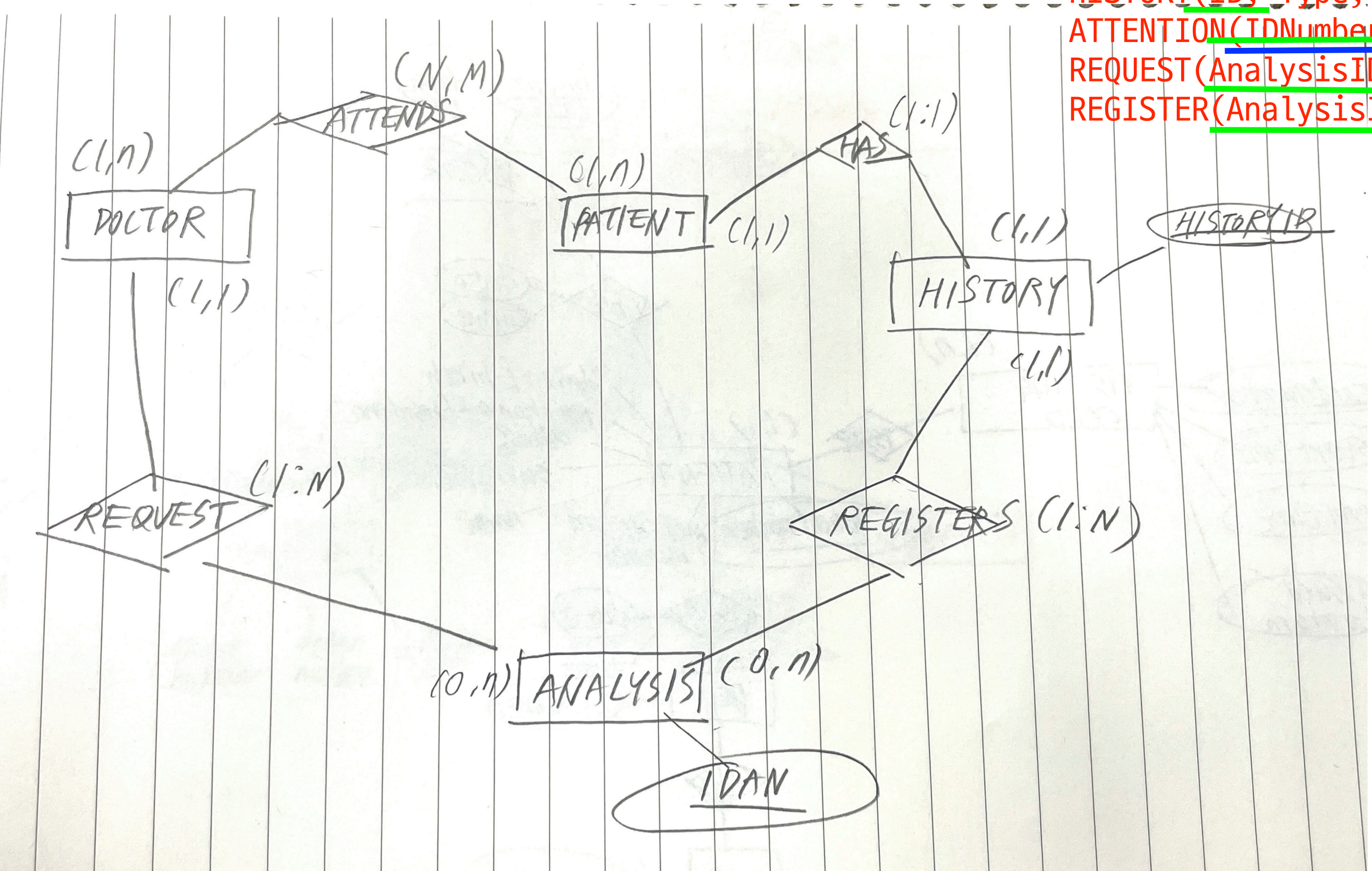
Exercises (III)

2.- Build an E/R diagram and obtain the corresponding Relational Model for the database of a hospital. In the database we want to collect information about the doctors who work there and the patients they treat.

1. For the doctors we know their ID number, name and registration number.
2. For the patients we know their ID, name and social security card number.
3. Each patient will have an assigned history where information about all the tests they have undergone in their life will be collected.
4. For each analysis we know the date it was carried out, the parameter analysed (sugar, leucocytes, psa, etc.) and its value. In each analysis only one value is analysed.
5. We want to know which doctor has requested the analysis.
6. A patient can be seen by more than one doctor.

DOCTOR(ID, RegistraitonNUm, Name)
PATIENT(ID, HealthCardNum, Name)
ANALYSIS(AnalysisID, IDNumberD, HistoryID)
HISTORY(ID, Type, Date, IDNumberP)
ATTENTION(IDNumberD, IdNumberP)
REQUEST(AnalysisID, IDNUmberD)
REGISTER(AnalysisID, HistoryID)

green: primary key
blue: foreign key



POR FAVOR,
CUIDEMOS LAS INSTALACIONES

SON UNA AYUDA
PARA NUESTRA FORMACIÓN

Exercises (III)

DOCTOR (DNI, RedistationNumber, Name)

PATIENT (DNI, HealthCardNumber, Name)

ANALYSIS (AnalisisID, Date, Value, Parameter, **DDNI**, **PDNI**)

ATTENTION (IDAttention, **DDNI**, **PDNI**)

The Relational Model. Transforming Relationships

N-ary relationships

According to LA, Look-Across, approach, the semantics of fig. 2 is as follows: a dealer company can distribute several products and a product can be distributed by several dealers (each pair of one dealer and one state can be associated to many products and each pair of one product and one dealer can be associated to many states). But in each state a product cannot be distributed by more than one dealer.

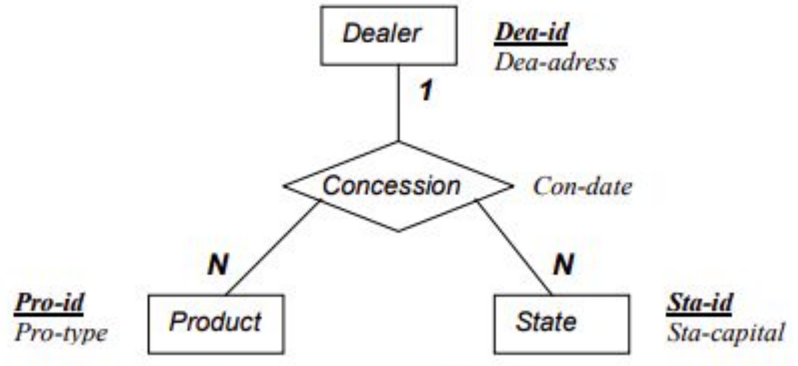


Fig.2: An example of **1:N:N** relationship

The Relational Model. Transforming Relationships

N-ary relationships

STATE (<u>Sta-id</u>, Sta-capital)

PRODUCT (<u>Pro-id</u>, Pro-type)

DEALER (<u>Dea-id</u>, Dea-address)

CONCESSION (**<u>Pro-id, Sta-id</u>, Dea-id**, Con-date)

Each one of the attributes Dea-id, Pro-id and Sta-id in the table Concession should be declared as not null foreign key.

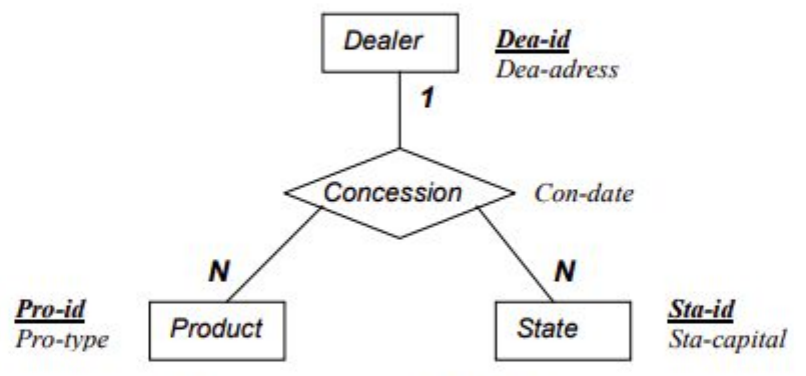
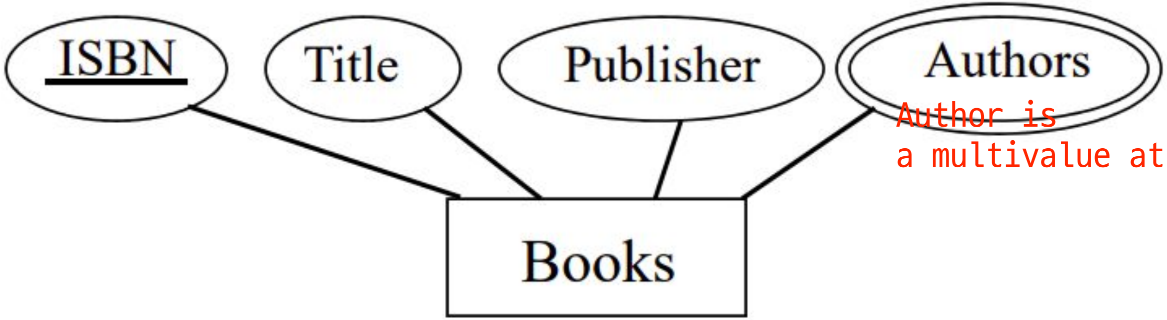


Fig.2: An example of ***1:N:N*** relationship

Multivalued Attributes

The Relational Model. Transforming multivalued attributes



Author is
a multivalue attribute

==> Books (ISBN, Title, Publisher)

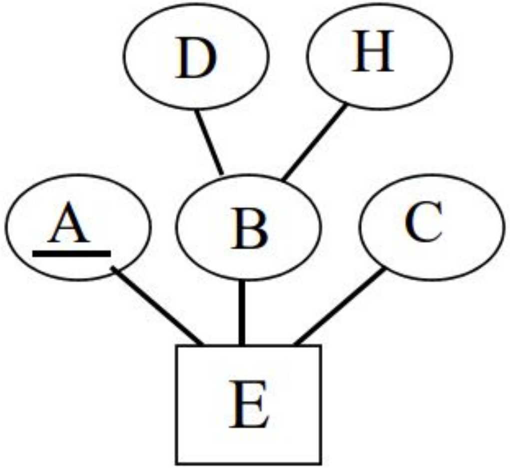
Book_Authors (ISBN, Author)

Book_Authors table
primary key: ISBN and Author
foreign key: ISBN

- Define Book_Authors.ISBN as a foreign key referencing Books.ISBN

The Relational Model. Transforming composite attributes

1. Use only simple attributes and ignore the composite attribute.

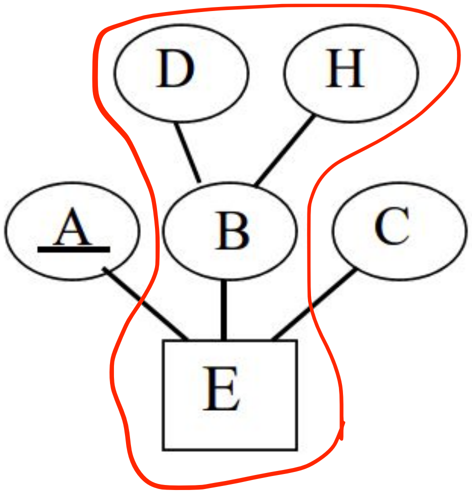


==>

E(A, D, H, C)

The Relational Model. Transforming composite attributes

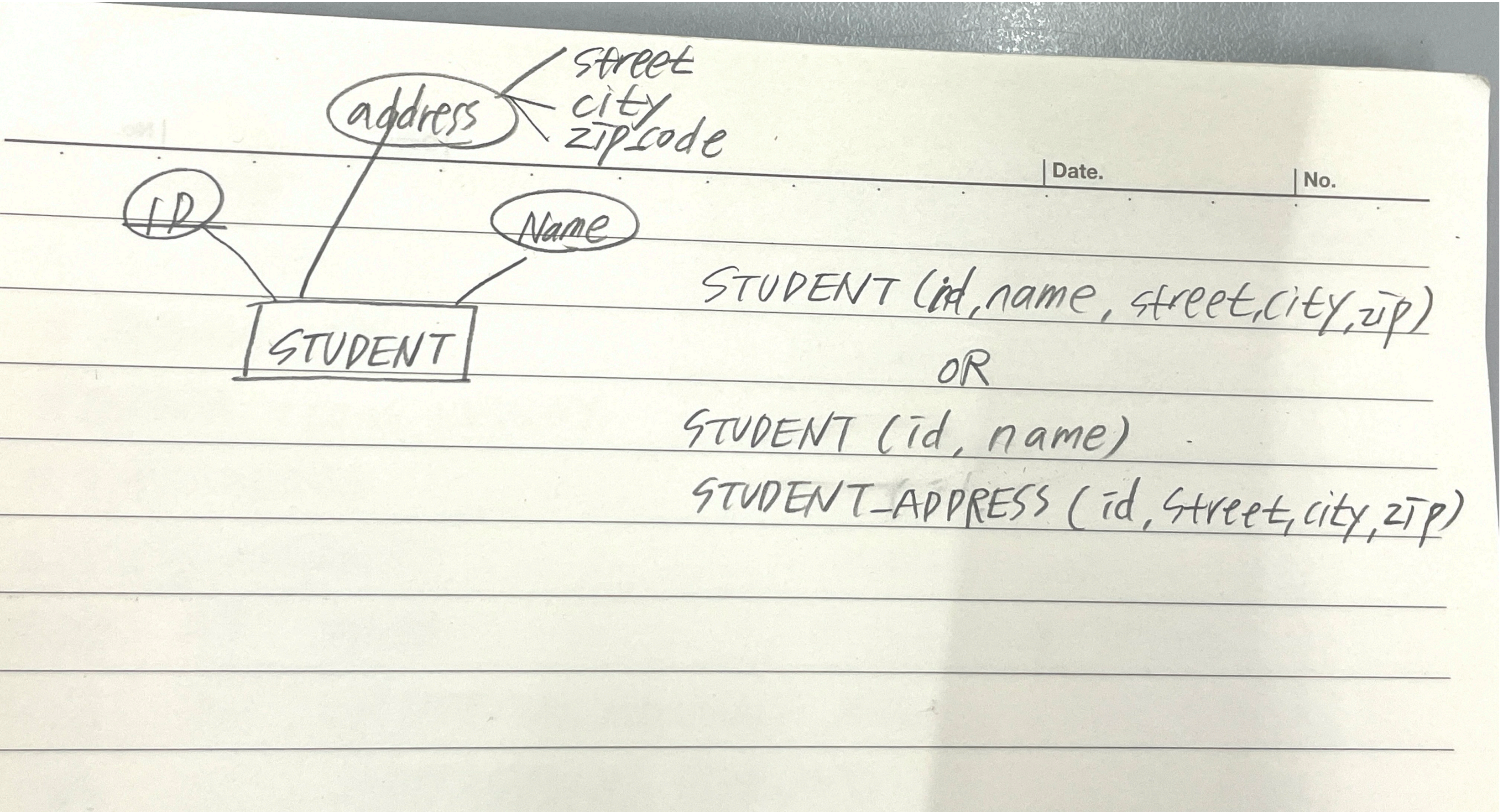
2. Transform the composite attribute to a separate relation.



==>

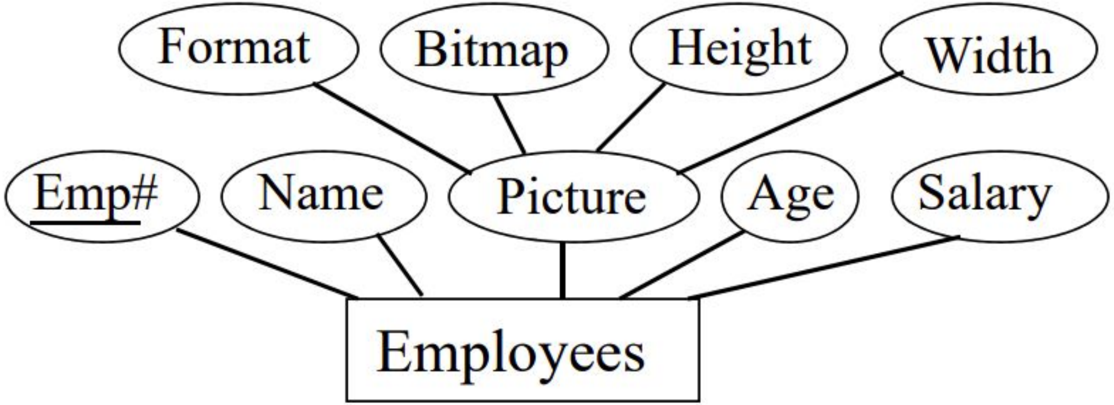
E(A, C),
E_B (A, D, H)

Example of Transforming composite attributes



The Relational Model. Transforming composite attributes

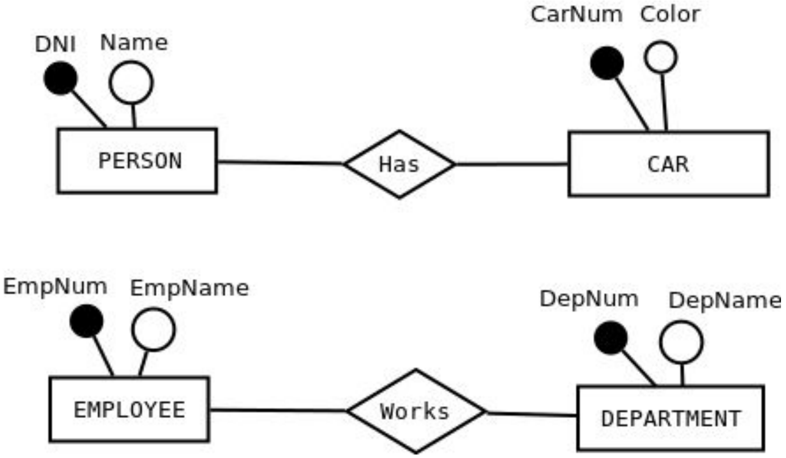
2. Transform the composite attribute to a separate relation.



Employees (Emp#, Name, Age, Salary)

Emp_Pic (Emp#, Bitmap, Format, Height, Width)

Make your own tips



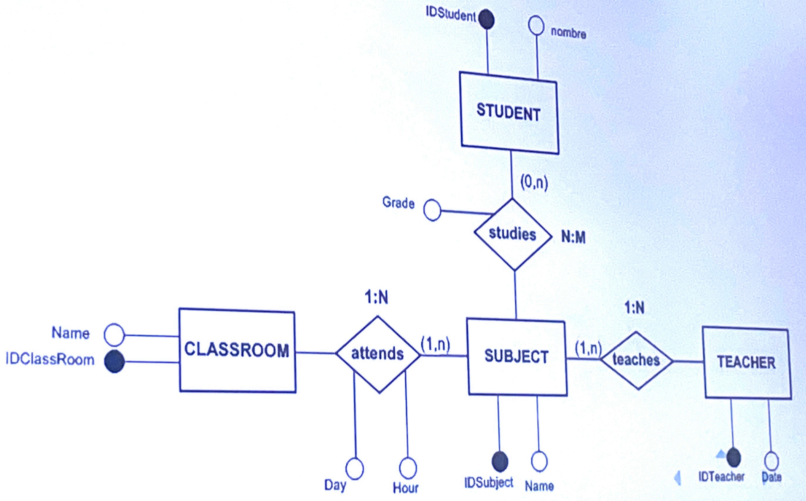
Exercises (III). University office

3.- Build an E/R diagram and obtain the corresponding Relational Model for the office of a university that keeps information on each of the subjects taught, the professors who teach them, the number of students enrolled, the day, time and classroom in which each subject is taught.

1. For each student and teacher, the ID and name are known.
2. The name of each subject is known.
3. A grade is recorded for each student-subject pair.
4. Each subject is taught by a single teacher.
5. The name of each classroom is known.
6. Several subjects are taught in one classroom. A subject can be taught on different days and at different times.

If multivalue is on the exam, she will use the word “multi”

Exercises (III)



Exercises (III)

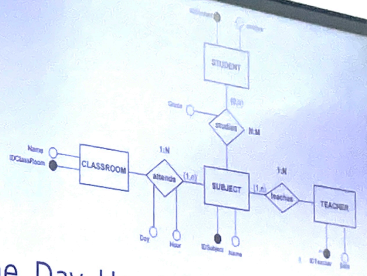
STUDENT (<u>DNI</u>, Name)

TEACHER (<u>DNI</u>, Name)

CLASSROOM (<u>IDClassRoom</u>, Name, Day, Hour)

SUBJECT (<u>IDSubject</u>, Name, **TDNI**, **IDClassRoom**)

STUDIES (<u>IDStudies</u>, **SDNI**, **IDSubject**, Grade)



The Relational Model. Transforming Relationships

To transform a reflexive relationship into the relational model, assume that it is a binary relationship with the particularity that the two entities are equal and apply the rules of the sections.

Reflexive Relationships

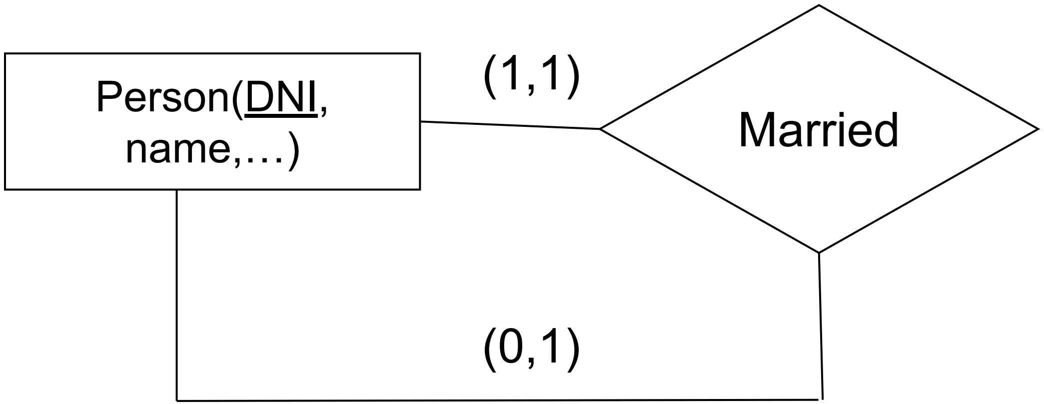
The Relational Model. Transforming Relationships

How can we transform the reflexive relationships?

Reflexive relations: Cardinality 1:1

- The relationship does not generate a table
- In the entity the **key** is inserted **twice** with different name, **only one of them will be key**

1:1



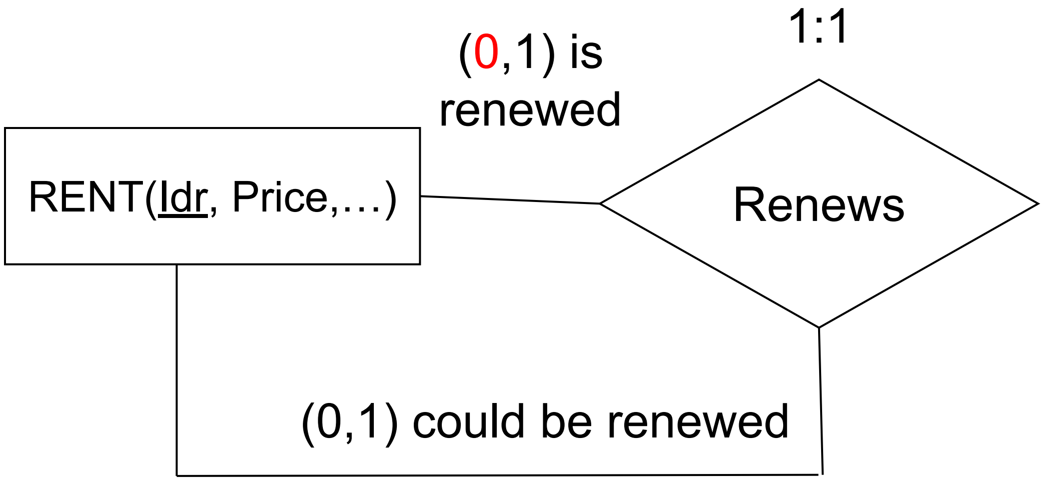
PERSON(<u>DNI</u>,**ParnerDNI**,name)

PartnerDNI is a foreign key pointing to the primary key DNI in the same table

The Relational Model. Transforming Relationships

Reflexive relations: Cardinality 1:1

- The relationship generate a table if both minimum cardinalities involved 0



RENT(<u>Idr</u>,Price)
RENEWS (**<u>Idr</u>**, **IdrOLD)**

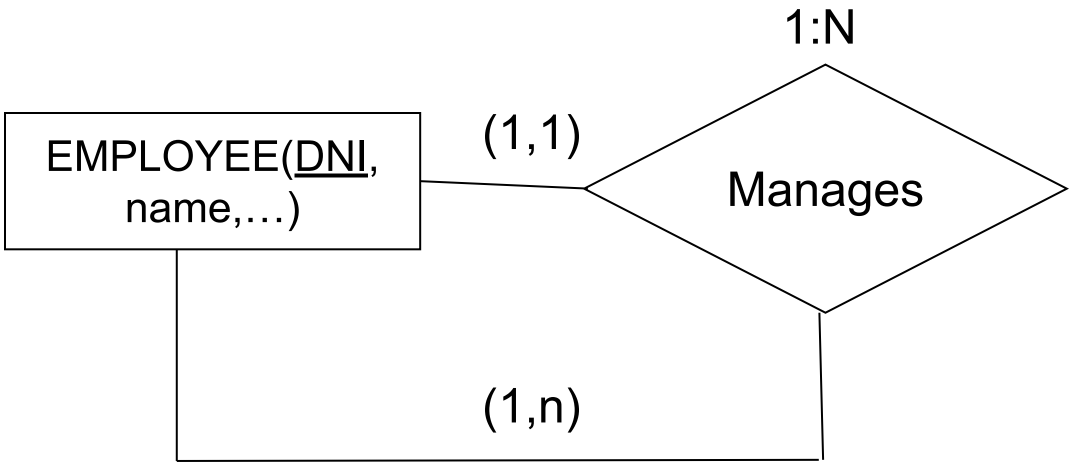
The Relational Model. Transforming Relationships

Reflexive relations: Cardinality 1:N

- The relationship generate a table if minimum cardinalities involved 0

an employee can manage one or several employees

If you are managed, you are managed by one employee

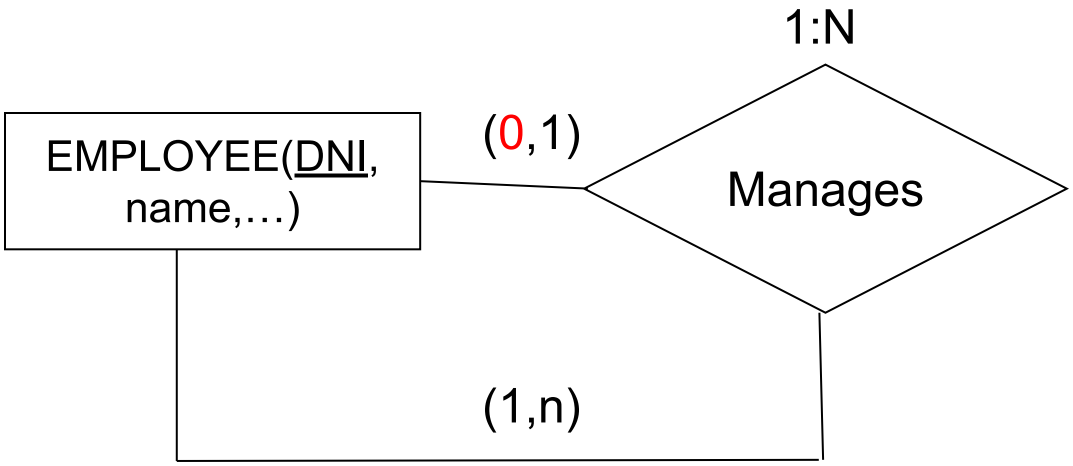


EMPLOYEE(DNI,name, **DNI2**)

The Relational Model. Transforming Relationships

Reflexive relations: Cardinality 1:N

- The relationship generate a table if minimum cardinalities involved 0

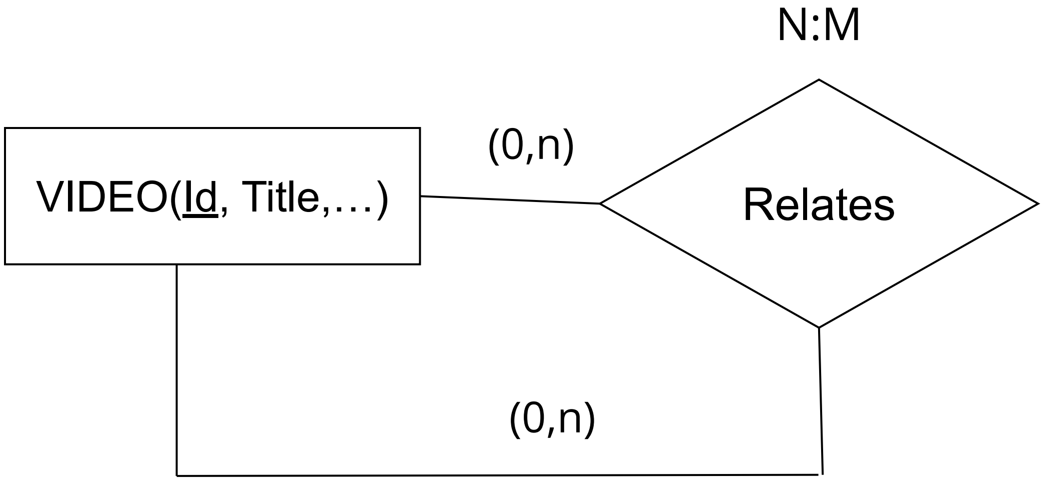


EMPLOYEE(DNI,name)
MANAGES(DNI,name, **DNI2**)

The Relational Model. Transforming Relationships

Reflexive relations: Cardinality N:M

- A table is created for the entity and another one for the relationship.
- The key of the relationship will compose of two attributes.



VIDEO(<u>Id</u>,Title)
RELATES(**<u>IdOriginal, IdRelated</u>**)

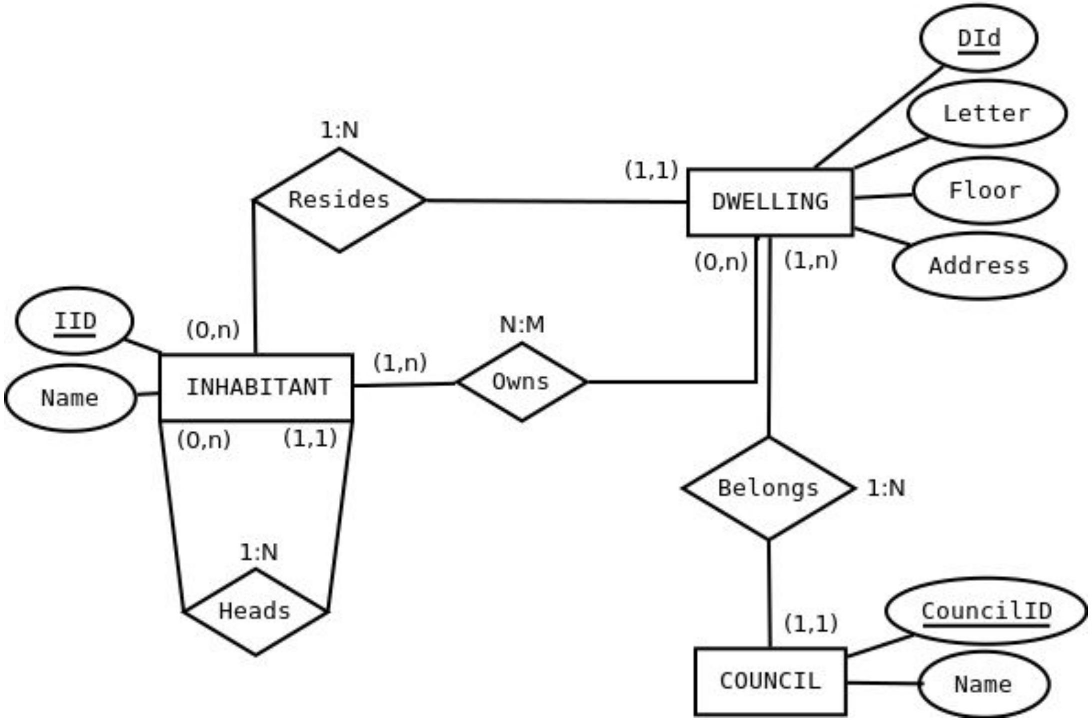
<u>IdOriginal , IdRelated</u> are both PK. Every original video is related to other video. A related video can also be related.

Exercises (III). Information system on councils

4.- Build an E/R diagram and obtain the corresponding Relational Model, which gathers the organisation of an information system on councils, dwellings/houses and inhabitants.

1. For each inhabitant, the ID number and name are known.
2. For each dwelling, the street, number, floor, letter are known.
3. For each council, the name is known.
4. A inhabitant can only live in one dwelling and reside in one municipality, but can own more than one dwelling.
5. It is interesting to reflect the relationship of each inhabitant with their head of household. An inhabitant may have only one head of household, but may be the head of household of many.
6. It is assumed that there are no councils without dwellings, but there may be councils without inhabitants.

Exercises-III



Exercises (III)

COUNCIL (<u>CouncilId</u>, Name)

INHABITANT (<u>IId</u>, Name, **DId, IIdHead_Household**)

DWELLING (<u>DId</u>, Address, Floor, Letter, **IDCouncil**)

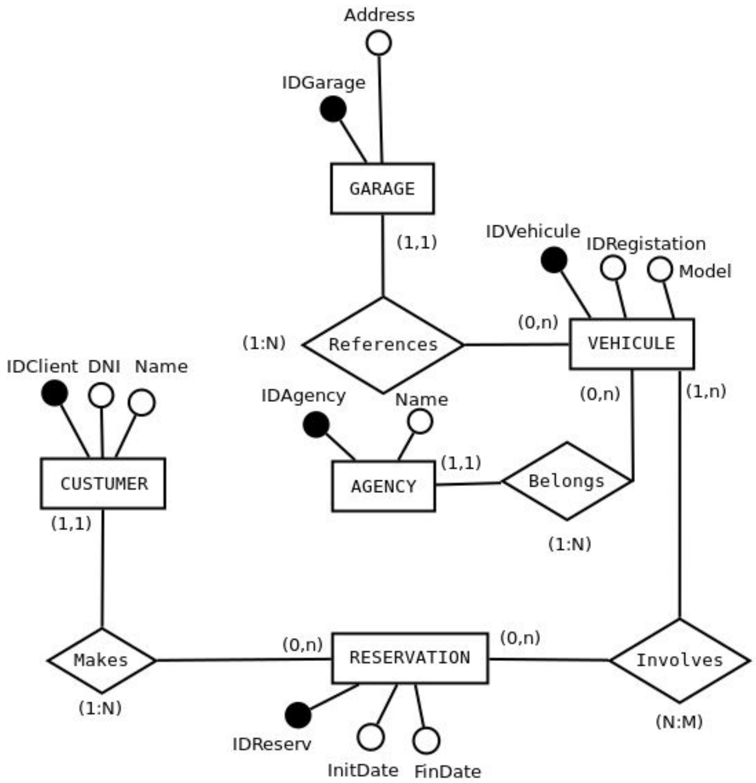
OWN (<u>**IId, Did**</u>)

Exercises (III). Car rental company

5.- Build an E/R diagram and obtain the corresponding Relational Model for a database that stores information about the reservations made by the clients of a car rental company. It must be taken into account:

1. For each customer, the ID number and name are known.
2. For each vehicle, the registration number and model are known.
3. A customer can have several reservations at any given time.
4. A reservation is made by a single customer, but can involve several cars.
5. Every car belongs to a certain agency.
6. We know the name of each agency.
7. Each car is assigned a reference garage that cannot be changed. For each garage we know the address.
8. In the database there may be customers who have not yet made a reservation.
9. All entities have an alphanumeric key that uniquely identifies them.
10. It is important to record the start and end date of the booking period.

Exercises (III)



Exercises (III)

CUSTOMER (IDClient, DNI, Name)

GARAGE (IDGarage, Address)

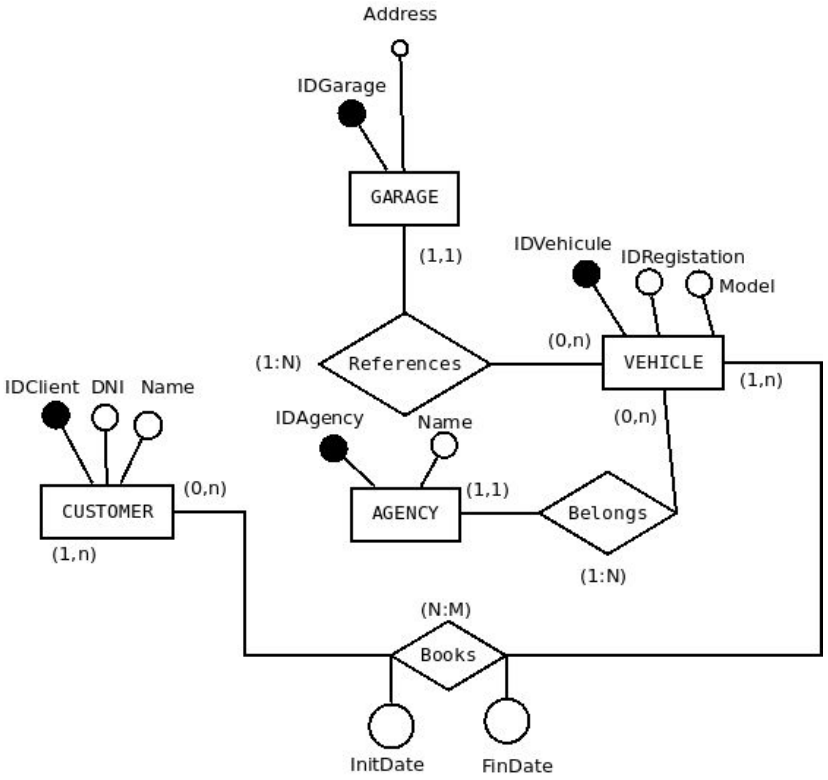
AGENCY (IDAgency, Name)

VEHICLE (IDVehicule, IDRegistation, Model, **IDGarage**, **IDAgency**)

RESERVATION (IDReserve, InitDate, FinDate, **IDClient**)

VEHICLERESERVATION (**IDVehicule**, **IDReserve**)

Exercises (III)



Exercises (III)

CUSTOMER (IDClient, DNI, Name)

GARAGE (IDGarage, Address)

AGENCY (IDAgency, Name)

VEHICLE (IDVehicle, IDRegistation, Model, **IDGarage**, **IDAgency**)

BOOKS (**IDClient, IDVehicle**, InitDate, FinDate)

The Relational Model. Transforming Relationships

Hierarchical relationships

Some guidelines can be given to serve as a reference, but in most cases, **it will depend on the specific problem. These guidelines are**:

- A table shall be created for the super type entity. Unless it has so few attributes that leaving it is a complication.
- If the subtype entity has no attributes and is not related, it disappears.
- If the subtype entity has its own attributes, a table is created. If it does not have its own key, it inherits the key of the supertype.
- If the relationship is **exclusive**, the attribute that generates the hierarchy is incorporated into the table of the supertype.
    - If a table has been created for each subtype it is not necessary to incorporate this attribute to the supertype entity.
- If the relationship is **inclusive**.
    - A table is created for each subtype.
    - A table is created for the supertype.
    - The attribute that generated the hierarchy is not incorporated in any table.

Hierchical Model

The Relational Model. Transforming Relationships

There are three possible answers for hierchical relationship

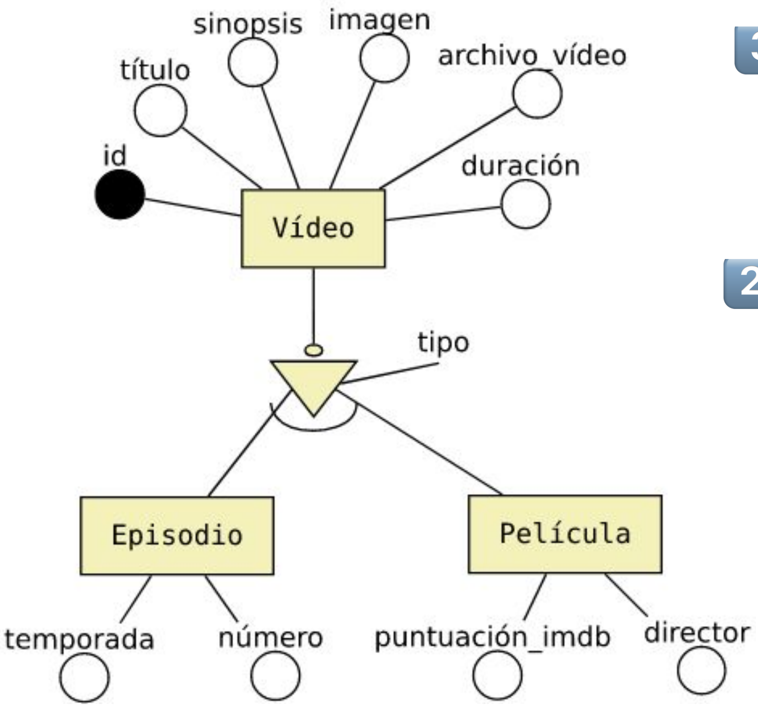
There are several solutions for the transition to tables of a specialisation. The solution chosen in each case will depend on the type of specialisation we are solving: total, partial, inclusive or exclusive.

The 3 possible solutions we can apply are the following:

1. Create a single table for the superclass. In this case all the attributes of the subclasses would be stored in the superclass.
2. Create a table only for the subclasses. In this case the attributes of the superclass would have to be stored in each of the subclasses.
3. Create a table for each of the entities, both for the superclass and the subclasses. In this case the subclasses would have to store the primary key of the superclass.

⭐ If it is a hierchical relationship, always add TIPO ⭐

The Relational Model. Transforming Relationships



3️⃣ Answer 3: Create a table for each entity.

Remember to put relational attribute in the super class

- VIDEO (<u>id</u>, title, sinopsis, image, archive, length, type)
- EPISODE (**<u>id</u>**, season, number) → **id**: FK de VIDEO (id)
- FILM (**<u>id</u>**, score, director) → **id**: FK de VIDEO (id)

2️⃣ recommend

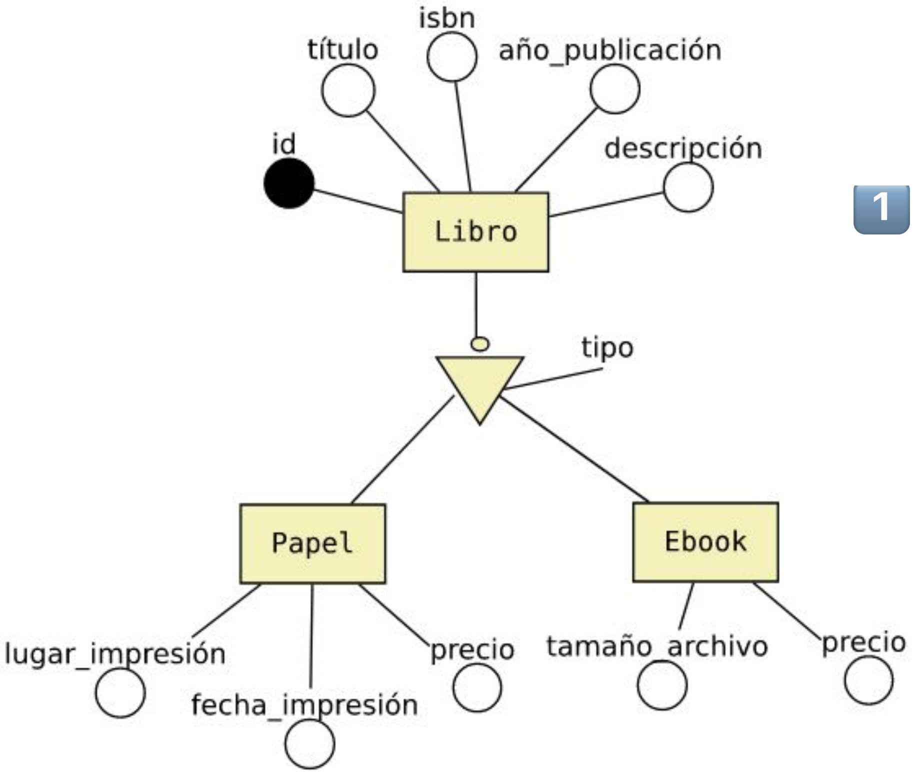
Answer 2: Create a table for subclasses only.

- EPISODE (<u>id</u>, title, sinopsis, image, archive, length, season, number)
- FILM (<u>id</u>, title, sinopsis, image, archive, length, score, director)

get all attribute from super class + own attribute

there is no foreign key

The Relational Model. Transforming Relationships



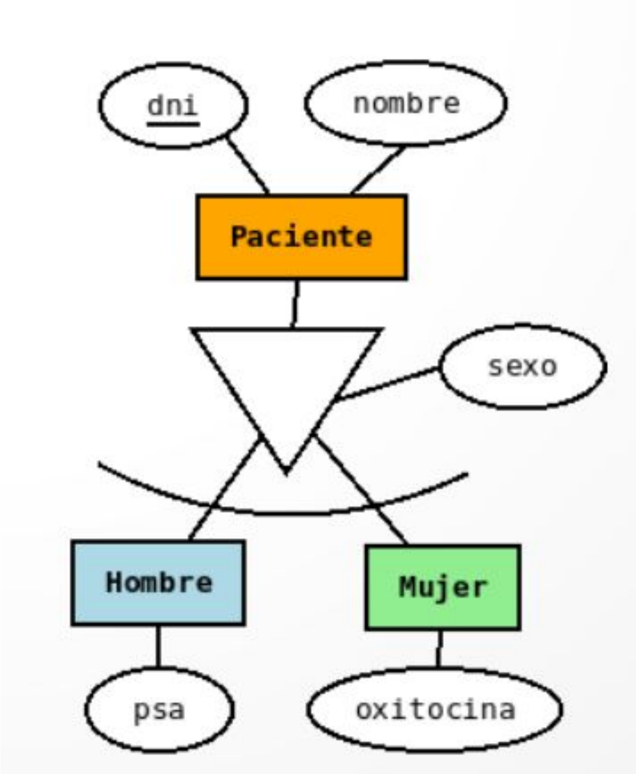
It is possible to use answer 3, although answers 2 or 3 are also possible.

1 when there are not many attribute

Answer 1. Create a single table for the superclass.

- BOOK (<u>id</u>, title, isbn, printing_year, description, type, printing_place, printing_date, paper_price, file_size,, ebook_price)

The Relational Model. Transforming Relationships

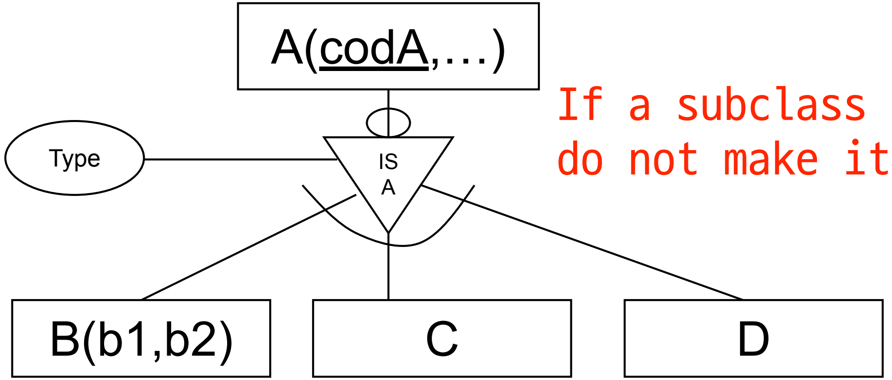


1. PATIEN (<u>dni</u>, name, sex, psa, oxytocin)

2. MANPATIEN (<u>dni</u>, name, psa)
WOMANPATIEN ((<u>dni</u>, name, oxytocin)

3. PATIEN (<u>dni</u>, name, sex)
MAN (**<u>dni</u>**, psa)
WOMAN (**<u>dni</u>**, oxytocin)

The Relational Model. Transforming Relationships



If a subclass do not have attributes, do not make it into table

A(codA,Type,...)

B(codA,b1,b2)

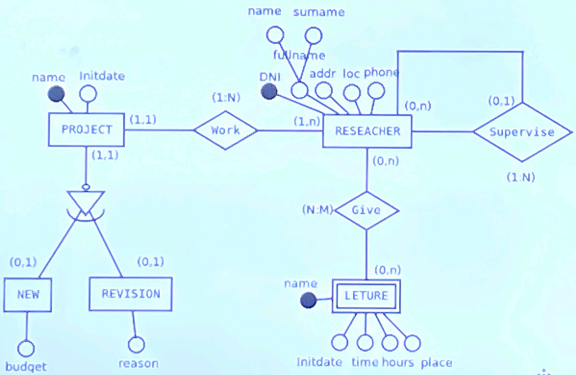
C and D are deleted if they don't have attributes, all their instances will be in table A.

Exercises (III). Statement: Research center

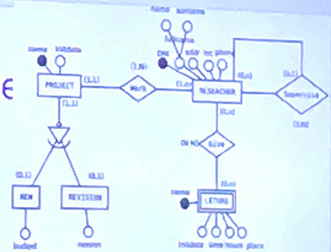
6.- A number of projects are carried out in a research centre. Each research project is carried out by a number of researchers. If there is a project, there is at least one researcher working on it. Each researcher works on only one project in the centre, the one assigned to him or her. For each project we are interested in its name (which is unique per project) and the date when the project started.

1. There are only 2 types of research projects: new and revision. For new projects, we are interested in recording the financial budget (in euros) available to carry out the project, while for revision projects we are interested in keeping a text explaining the reason for the revision (e.g. "Initial calculation error" or "Adaptation to new market needs").
2. There are managers and non-managerial researchers among the researchers. Each non-manager researcher is supervised by a manager, while managers do not have a superior manager to supervise them. For each researcher we would be interested in recording his/her full name (although separated into first and last names), ID number, address, location, and telephone number.
3. In addition, researchers will give lectures at other centres on their research, although not all researchers will do so. Each lecture will be given by one or more researchers. The most gifted researchers may even participate in more than one conference. For each lecture we are interested in the name, date and time of the start of the lecture, number of hours of the lecture and the place where the lecture will be held (e.g. at the Faculty of Statistics).

Exercises (III). CDM - ERD



Exercises (III). LDM - The



PROJECT (<u>PrName</u>, InitDate, type)

NEW (**<u>PrName</u>**, Budget)

REVISION (**<u>PrName</u>**, Reason)

RESEARCHER (<u>DNI</u>, **PrName**, ResName, ResSurname, Address, Loc, Tel)

LECTURE (**<u>DNIR,</u>** <u>LecName</u>, InitDate, Time, Place, Hours) → **DNIR** is **FK** to RESEARCHER

GIVES (**<u>DNI, LecName</u>**)

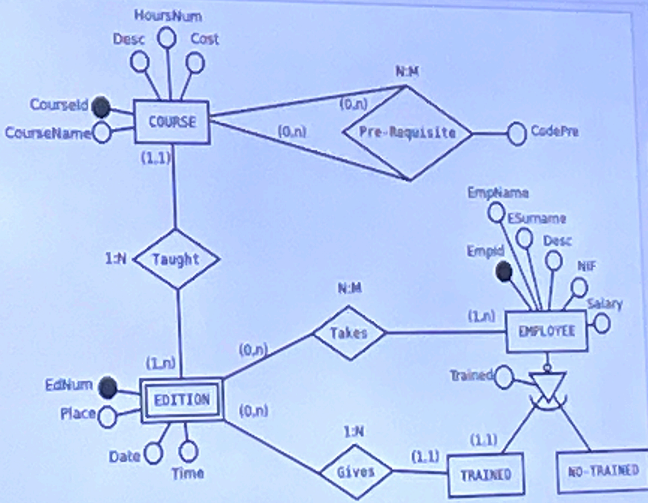
SUPERVISOR (**<u>DNIsupervisor, DNIsupervised</u>**)

Exercises (III). Statement: Training courses

7.- The training department of a company wants to build a database to plan and manage the training of its employees.

1. The company organises internal training courses for which it wants to know the course code, the name, a description, the number of hours and the cost of the course.
2. A course may have as a prerequisite the completion of other course(s), and in turn the completion of one course may be a prerequisite for other courses. A course that is a prerequisite for another course may be mandatory or only recommended.
3. The same course has different editions, i.e. it is held in different places, on different dates and at different times (intensive, morning or afternoon). Only one edition of a course can be held on the same starting date.
4. The courses are given by the company's own staff.
5. The employee's employee code, name and surname, address, telephone number, tax identification number, date of birth, nationality, sex, signature and salary, as well as whether or not he/she is qualified to teach courses, can be stored.
6. The same employee can be a teacher in one edition of a course and a student in another edition, but can never be both at the same time (in the same edition he/she either teaches it or receives it).

Exercises (III). CDM - ERD



Exercises (III). LDM - The Relational Model

COURSE (<u>CourseId</u>, CourseName, Desc, HoursNum, Price)

EDITION (<u>**CourseId**, EdNum</u>, Date, Place, Time, **EmpId**)

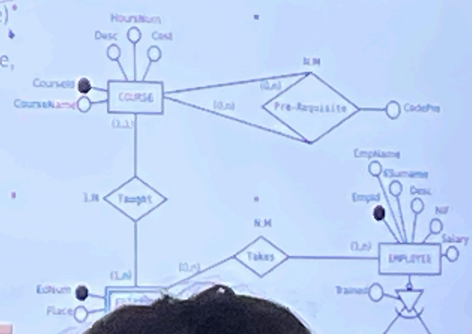
PRE-REQUISITE (<u>**HasPre**, **IsPre**</u>, CodePre)*

EMPLOYEE (<u>EmpId</u>, EmpName, Surname, Trained*)

TAKES (<u>**EmpId**, **CourseId**, **EdNum**</u>)

TRAINED (**EmpId**)

NO_TRAINED (**EmpId**)



The Relational Model. Conditions

- The rows/tuplas/records of a DB schema must fulfil/satisfy a number of conditions over time.
- Not everything is representable in an entity/relationship model.
    - Example: for a given relationship that the value of one attribute is less than the value of another.
    - Example: We cannot establish that a basketball team can only have 5 players on the court at any given time.

The Relational Model. Steps

- Identify entities and attributes.
- Identify keys.
- Establish relationships.
- Eliminate loops.
- Try to obtain the minimum number of tables to avoid redundancies.
- Textual specification of integrity conditions.

References

- https://gestionbasesdatos.readthedocs.io/es/latest/Tema2/Teoria.html#fase-2-del-diseno-diseno-logico-modelo-relacional